

KQL Attack Detection Cheat Sheet

23 common attack techniques across the kill chain, with detection logic and example KQL queries.

Kill Chain Stage	Attack / Technique	MITRE ID	Detection Logic	KQL Example
Initial Access	Brute Force Authentication	T1110	Repeated failed logons against a single account in a short window.	SecurityEvent where EventID==4625 summarize Fails=count() by TargetAccount, bin(TimeGenerated,10m) where Fails>10
Initial Access	Password Spray	T1110.003	One source IP failing logon across many distinct accounts.	SigninLogs where ResultType!=0 summarize Accts=dcount(UserPrincipalName) by IPAddress, bin(TimeGenerated,10m) where Accts>10
Execution	Suspicious PowerShell	T1059.001	Encoded, hidden-window, or download-cradle PowerShell commands.	DeviceProcessEvents where FileName=="powershell.exe" and ProcessCommandLine has_any (" -enc", "-EncodedCommand", "IEX", "DownloadString", "-W Hidden")
Execution	Suspicious Cmd Execution	T1059.003	cmd.exe spawning reconnaissance or unusual child processes.	DeviceProcessEvents where InitiatingProcessFileName=="cmd.exe" and FileName in~ ("whoami.exe", "net.exe", "nslookup.exe")
Execution	LOLBAS Abuse	T1218 / T1105	Signed native binaries invoked with an HTTP reference (download).	DeviceProcessEvents where FileName in~ ("certutil.exe", "bitsadmin.exe", "mshta.exe", "regsvr32.exe") and ProcessCommandLine has_any ("http://", "https://")
Execution	Ingress Tool Transfer	T1105	A download command resulted in a new file written to disk.	DeviceFileEvents where InitiatingProcessFileName in~ ("powershell.exe", "certutil.exe") and FolderPath has_any (@"\Windows\", @"\Temp\")
Persistence	Registry Run Key	T1547.001	New value written to a common autorun registry path.	DeviceRegistryEvents where RegistryKey has @"CurrentVersion\Run"
Persistence	Scheduled Task Creation	T1053.005	New scheduled task created via schtasks.	DeviceProcessEvents where FileName=="schtasks.exe" and ProcessCommandLine has "/create"
Persistence	New Service Creation	T1543.003	New service installed, especially with a short/random name.	SecurityEvent where EventID==7045 where ServiceName matches regex @"^[A-Za-z]{4,8}\$"
Defense Evasion	Clearing Event Logs	T1070.001	wevtutil used to clear Security/System/Application logs.	DeviceProcessEvents where ProcessCommandLine has "wevtutil" and ProcessCommandLine has " cl "
Defense Evasion	Disabling Security Tools	T1562.001	PowerShell command disabling Defender real-time protection.	DeviceProcessEvents where ProcessCommandLine has_any ("Set-MpPreference", "DisableRealtimeMonitoring")
Defense Evasion	Masquerading	T1036	Well-known process name running from an unexpected file path.	DeviceProcessEvents where FileName=="svchost.exe" and FolderPath !has @"\Windows\System32"
Credential Access	LSASS Credential Dumping	T1003.001	A process opens a handle to LSASS memory (e.g. Mimikatz-style tools).	Sysmon EventID=10 where TargetImage endswith "lsass.exe"

Kill Chain Stage	Attack / Technique	MITRE ID	Detection Logic	KQL Example
Credential Access	Kerberoasting	T1558.003	Kerberos service ticket requested using weak (RC4) encryption.	SecurityEvent where EventID==4769 and TicketEncryptionType=="0x17"
Lateral Movement	PsExec-Style Service Creation	T1021.002	New service pointing to an executable with a randomized name.	SecurityEvent where EventID==7045 where ImagePath matches regex @"[A-Za-z]{8}\.exe\$"
Lateral Movement	Named Pipe Lateral Movement	T1021.002	Named pipes matching known remote execution tool naming.	Sysmon EventID=17 or 18 where PipeName has_any ("RemCom", "PSEXESVC", "status")
Lateral Movement	WMI Lateral Movement	T1047	wmic.exe used to execute commands against a remote host.	DeviceProcessEvents where FileName=="wmic.exe" and ProcessCommandLine has "/node:"
Lateral Movement	Pass-the-Hash	T1550.002	NTLM network logon that skips normal interactive authentication.	SecurityEvent where EventID==4624 and LogonType==9 and AuthenticationPackageName=="NTLM"
Lateral Movement	RDP Lateral Movement	T1021.001	Interactive RDP logon (Type 10) to a host.	SecurityEvent where EventID==4624 and LogonType==10
Command and Control	C2 Beacons	T1071	Host making periodic connections to the same external IP.	DeviceNetworkEvents summarize Conns=count() by RemoteIP, bin(Timestamp,1h) where Conns>50
Command and Control	DNS Tunneling	T1071.004	Unusually high volume of DNS queries from a single host.	DeviceNetworkEvents where RemotePort==53 summarize Q=count() by DeviceName, bin(Timestamp,1h) where Q>200
Exfiltration	Large Outbound Transfer	T1041	Abnormally large outbound data volume from a single host.	DeviceNetworkEvents summarize Bytes=sum(SentBytes) by DeviceName, bin(Timestamp,1h) where Bytes>500000000
Domain / AD	DCSync Attack	T1003.006	Directory replication request from a non-domain-controller account.	SecurityEvent where EventID==4662 and Properties has "1131f6aa-9c07-11d1-f79f-00c04fc2dcd2"

Note: Queries are written generically across Sentinel/Defender-style tables (DeviceProcessEvents, DeviceNetworkEvents, SecurityEvent) and Sysmon event IDs. Field and table names vary by data source — adjust to match your actual schema. Thresholds (e.g. Fails>10, Conns>50) are starting points, not fixed rules; tune them against your own environment's normal baseline.